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Application of Artificial Intelligence for Predicting Market Trends

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Abbreviation

API	Application Programming Interface
ANN	Artificial Neural Networks
ARIMA	AutoRegressive Integrated Moving Average
CNN	Convolutional Neural Networks
DL	Deep Learning
EDA	Exploratory Data Analysis
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
OHLC	(Open, High, Low, Closing)
RF	Random Forest
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Networks
SVR	Support Vector Regression

1 Introduction

With the rise of technological advancements such as global digitization, market prediction has entered a new era of innovation, transforming traditional trading models. This evolution is particularly significant in sectors like the automotive market, where accurate predictions drive critical decision-making for manufacturers, suppliers, and investors. Stock trading, fueled by continuous growth in market capitalization, has become a key focus for financial investors. To support this, analysts and researchers have developed sophisticated tools and techniques that forecast price movements, aiding in strategic decision-making. (Rouf, et al.)

1.1 Objectives of the Project

The volatility of stock prices, often influenced by national policies and social factors, presents significant challenges for investors seeking profitable returns in the stock market. This study uses Volkswagen, Mercedes-Benz, Porsche and BMW—four leading competitors in the global automotive industry—as a case study to analyze market dynamics and identify the least risky and most suitable automotive-maker for predicting future stock trends. To achieve this, we developed a forecasting model utilizing Long Short-Term Memory (LSTM) neural networks to predict stock market trends. Using historical data from 2015 to 2024, sourced from Yahoo Finance, we forecast stock movements over a one-month horizon. Our focus is on predicting key stock price components, including Open, High, Low, and Close (OHLC) prices. This study demonstrates the potential of advanced predictive models in enhancing financial decision-making and navigating the uncertainties of volatile markets.

1.2 Problem Statement

This paper investigates the research question of how artificial intelligence can be used to predict market developments. Various AI models and their areas of application are examined in order to make a well-founded statement about their predictive ability (Nabipour, Nayyeri, & Mosavi, 2020).

This paper investigates the research question of:

- how can artificial intelligence be used to predict market developments and why LSTM is suitable for stock forecasting?
- What are the optimal hyperparameters for LSTM to forecast stock price trends effectively for OHLC (Open, High, Low, Closing) prices?

- which automotive-maker industry is less risky and less volatile for more accurate LSTM results?
- Can predictive models based on historical stock price trends effectively predict future prices?
- What metrics (e.g., RMSE, MAE, MAPE) best evaluate the accuracy and reliability of stock price prediction models?

1.3 Structure of the Report

Figure 1 presents the outline of this study.

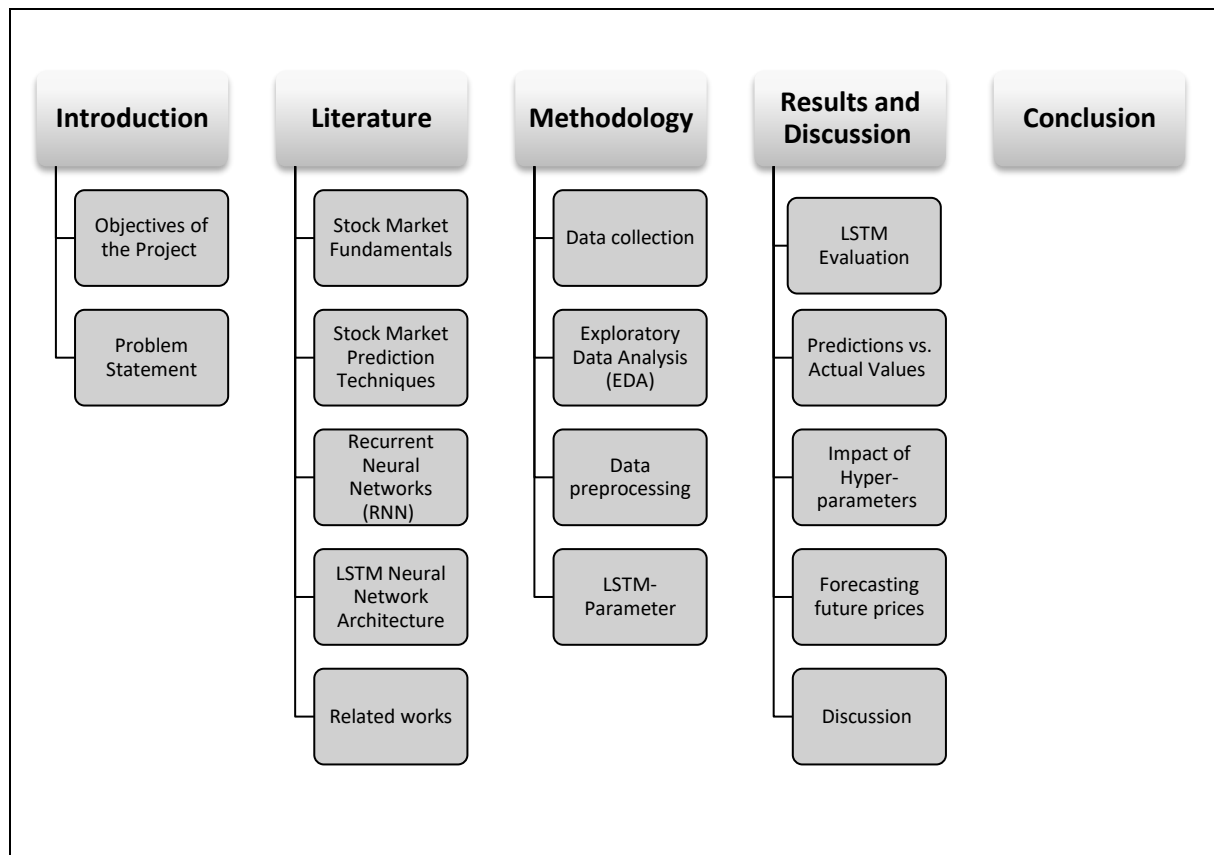


Figure 1: The outlines of this study.

2 Literature Review

2.1 Fundamentals of Stock Market

2.1.1 Stock Market

The stock market serves as a marketplace where individuals buy and sell shares of publicly traded companies, enabling businesses to raise capital for expansions, research, and development. This dynamic ecosystem facilitates wealth generation for both investors and corporations, offering opportunities for financial growth through strategic investments.

Stock prices fluctuate due to various factors, including economic indicators, corporate performance, geopolitical events, and market sentiment. These interdependent variables make stock trading a complex puzzle with multiple influencing pieces (Zakhidiv, 2024) .

The origins of the stock market trace back to the 16th-century Amsterdam Stock Exchange, where the Dutch East India Company became the first publicly traded company, laying the foundation for modern stock trading. Over centuries, the stock market has evolved into a robust global platform for physical and digital trade in company shares (Oscar & Joost, September 2004).

While the stock market offers opportunities for wealth creation, it also involves significant risks. This was evident during the 2008 Financial Crisis, when the U.S. housing bubble burst, triggering a global economic downturn and substantial investor losses. As shown in the chart below, the market plummeted sharply in 2008, with the global value falling from **60.46 trillion** in 2007 to **32.42 trillion** in 2008. Despite such severe declines, markets have historically recovered through policy interventions designed to stabilize economies (Shiller, 2008).

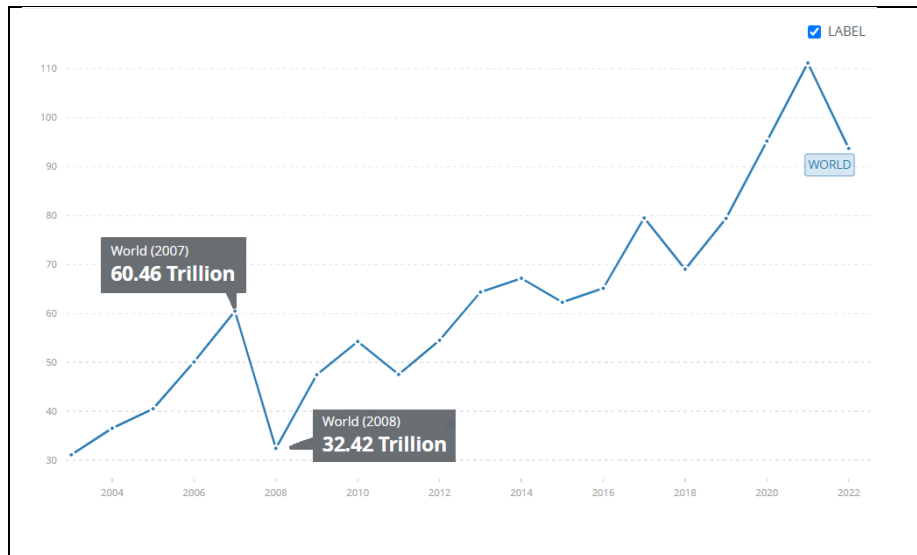


Figure 2: Trends in Global Market Value (2004–2022) with Focus on the 2008 Financial Crisis (Data Worldbank).

Successful stock trading requires adopting sound strategies to navigate the market's risks and opportunities. Modern forecasting techniques, particularly time series models, have become pivotal in predicting stock price movements and improving financial decision-making.

2.2 Stock Market Prediction Techniques

2.2.1 Traditional techniques

Traditional stock market prediction techniques, such as regression analysis, econometric modeling, and time series analysis, provide valuable tools for understanding market dynamics and forecasting trends. Regression examines the influence of factors like company earnings or GDP growth on stock prices, while econometric models combine economic theory with statistical methods to estimate causal relationships, such as how interest rates affect investments (Zaxidov, January 2024). Time series analysis, including the widely used ARIMA model, leverages historical data for short-term forecasting. ARIMA, known for its robustness with linear data, excels in predicting stock indices by ensuring stationarity and optimizing parameters, often outperforming more complex models in short-term predictions (Adebiyi, Adewumi, & Ayo, 2014).

2.2.2 Modern techniques

Machine learning (ML) and deep learning (DL) methodologies have shown superior performance in stock price forecasting compared to traditional statistical methods due to their ability to handle complex, non-linear patterns and large datasets (Sirisha, Belavagi, & Attigeri, 2022).

ML techniques are now being heavily explored in stock price prediction, such as support vector regression (SVR), decision tree regression, and random forest (RF), are widely applied in stock prediction (Sirisha, Belavagi, & Attigeri, 2022). DL methods, including artificial neural networks (ANN), convolutional neural networks (CNN), and recurrent neural networks (RNN), have been extensively utilized in stock market prediction (Shahi, Shrestha, Neupane, & Guo, 2020). While ANN and CNN are powerful for various applications, they do not account for temporal dependencies in data, making RNN algorithms particularly valuable in this domain. Among RNNs, long short-term memory (LSTM) networks stand out for their ability to capture long-term dependencies and detect subtle patterns in sequential financial data. By addressing challenges such as vanishing and exploding gradients (Sherstinsky, 2020) LSTMs are especially effective in modeling the intricate trends and dependencies inherent in stock market behavior (Ma, Han, & Wang, 2021).

2.3 Recurrent Neural Networks (RNN)

For applications like stock prediction, recalling previous patterns is essential, as past trends often repeat. An artificial neural network with a sequential information structure is known as a Recurrent Neural Network (RNN). They are referred to as recurrent because they execute the same function on each sequence element, with the outcome depending on prior calculations. RNNs are loop based networks that enable data preservation. Each recurrent neural network comprises a chain of repeating neural network modules. A recurrent neural network includes loops, enabling information to be retained in the network. In Fig.1, a simple recurrent neural network with loops is shown. The neural network in Fig.1, A examines the input x_t and then generates output h_t . A loop enables data to be transferred from one network phase to the next (imtiaz & qusay, may 2022).

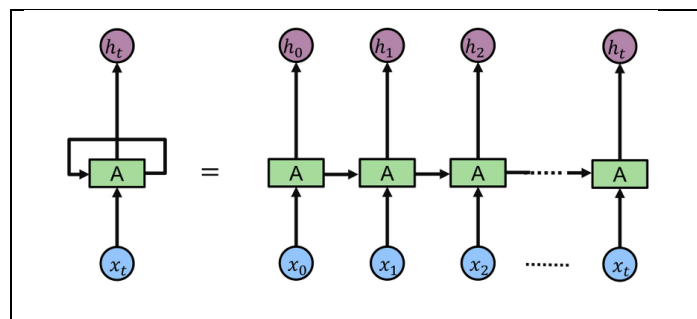


Figure 3: Simple recurrent neural network with loops characteristic (imtiaz & qusay, may 2022)

While RNNs are theoretically capable of handling long sequences, practical challenges such as vanishing and exploding gradients during backpropagation limit their ability to learn complex, long-term dependencies. **Vanishing gradients** occur when activation functions like Sigmoid or Tanh produce gradients between 0 and 1, causing values to shrink exponentially as they propagate backward through the network. **Exploding gradients**, on the other hand, result from excessively large weights amplifying gradients, leading to instability unless techniques like normalization or gradient clipping are applied. Backpropagation used to update model weights, works by minimizing the loss function, which quantifies prediction errors. Using the chain rule, gradients are calculated layer by layer, moving backward from the output to the input layer. This process helps refine weights to improve accuracy but is vulnerable to the aforementioned gradient issue (Sherstinsky, 2020). To overcome these challenges, advanced architectures like Long Short-Term Memory (LSTM) networks are commonly and successfully employed for stock market forecasting (Fischer & Krauss, 2018). LSTMs use gating mechanisms, such as input, forget, and output gates, to regulate the flow of information, enabling the network to effectively capture long-term dependencies and model intricate sequential patterns. These architectures address vanishing gradient issues and manage gradient size better, making them particularly effective for tasks like stock prediction, where context and memory are crucial (Dey, et al., August 2022).

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2.4 Long Short-Term Memory LSTM

Hochreiter and Schmidhuber (1997) introduced Long Short-Term Memory (LSTM) networks to address the long-term dependencies and vanishing gradient problem in recurrent neural networks (RNNs). LSTMs effectively capture long-term dependencies, making them well-suited for time series predictions. However, the issue of exploding gradients can still occur, which is why normalization techniques are often applied to stabilize training by scaling features equally (see Section 3.3.2). LSTMs are robust due to key components: cell state (memory), input gates

(store information), forget gates (discard information), and output gates (control output to the next layer or time step). And due to its parameters.

2.4.1 LSTM Components

LSTM consists of repetitive and complex network modules. An LSTM layer contains one or more recurrently connected blocks, known as memory blocks; information is remembered by memory blocks through many time-steps learning (Graves & Schmidhuber, 2005)

LSTMs have a structure similar to RNNs, with a chain-like configuration of repeating modules. However, unlike RNNs, each module in an LSTM consists of four distinct neural network layers that interact in a specialized way. These layers perform calculations through three key gates—**forget gate**, **input gate**, and **output gate**—as well as the **cell state**, which acts as the memory of the network. Table 1 provides a list of symbols to clarify the concepts discussed in the following sections.

Symbol	Meaning	Symbol	Meaning
f_t	forgot gate	h_{t-1}	output of previous LSTM block
i_t	input gate	x_t	input at current timestamp
o_t	output gate	b_x	biases for the respective gate (x)
σ	sigmoid function	C_{t-1}	LSTM previous memory content
$\tanh$	hyperbolic tangent function	$\tilde{C}_t$	LSTM current memory content
w_x	weight for respective gate (x) neurons	C_t	LSTM new memory content

Tabel 1: List of symbols used in LSTM model (imtiaaz & qusay, may 2022)

2.4.1.1 Forget gate layer

The first step in an LSTM is determining which information from the **cell state** (see section 3.4.2.3) should be discarded. This decision is made by the forget gate, a sigmoid layer that evaluates the previous output (h_{t-1}) and the current input (x_t). The forget gate generates a value between 0 and 1 for each element in the **cell state** (C_{t-1}), where 1 indicates "completely keep this information" and 0 means "completely forget i_t ." The previous output (h_{t-1}) serves as short-term memory, influencing the values in the cell state (Saurav, Avesh, Sharma, & Hossain, 2023) (imtiaaz & qusay, may 2022).

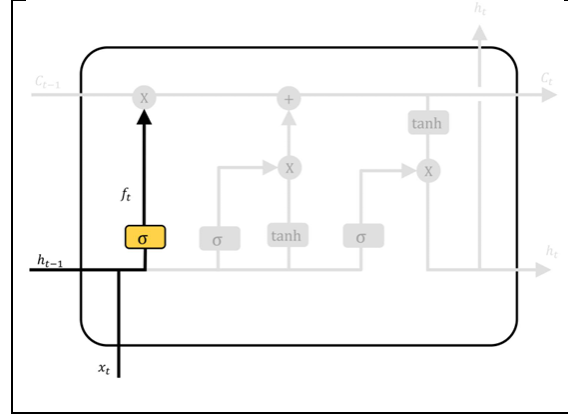


Figure 4: Information processed at Forget Gate in an internal cell (imtiaaz & qusay, may 2022).

As illustrated in Figure 4, the process begins by combining x_t and h_{t-1} to form a new vector, which is passed through the forget gate. This gate determines whether specific information is retained or removed from the cell state (C_t). The forget gate, denoted as f_t , uses a sigmoid function to regulate the degree of retention in long-term memory (Saurav, Avesh, Sharma, & Hossain, 2023) (imtiaaz & qusay, may 2022). Its operation is mathematically represented by Equation:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

2.4.1.2 Input Gate Layer

Following the first step, where the forget gate determines how much old information to discard, the second step in the LSTM process involves deciding what new information to store in the cell state. This decision is made by the **input gate layer**, as illustrated in Figure 5 (Saurav, Avesh, Sharma, & Hossain, 2023) (imtiaaz & qusay, may 2022).

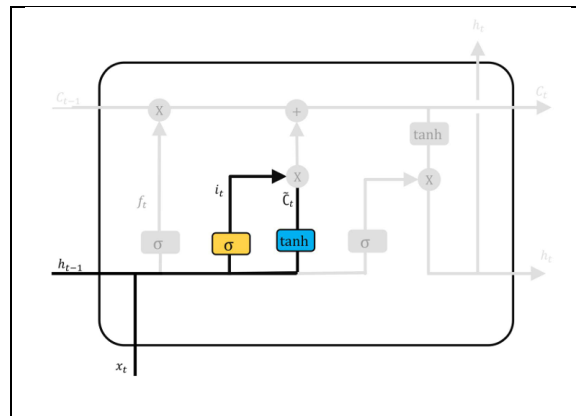


Figure 5: Information processed at Input Gate in an internal cell (imtiaaz & qusay, may 2022).

In this phase, the current input (x_t) and the previous output (h_{t-1}) are processed in two parts. The first part, the **input gate** (denoted as i_t), uses a sigmoid function to decide which values

should be updated. The operation of the input gate is described by Equation (2), and its function is shown in Figure 5. The second part, the **new memory candidate** (denoted as $\tilde{C}_t$), is controlled by a tanh function, which generates a vector of potential new values to add to the cell state. This operation is represented by Equation (3). Together, these components determine how much new information will be incorporated into the cell state (Saurav, Avesh, Sharma, & Hossain, 2023) (imtiaz & qusay, may 2022).

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (3)$$

2.4.1.3 cell state

The core component of an LSTM is the cell state (C_t), represented as a horizontal line running across the network. It serves as long-term memory, analogous to a conveyor belt that flows through the network, retaining and transferring information while undergoing regulated updates. The cell state is modified by interactions with the gates—special components that control what information is added, retained, or discarded (Saurav, Avesh, Sharma, & Hossain, 2023) (imtiaz & qusay, may 2022), as shown in Figure 6.

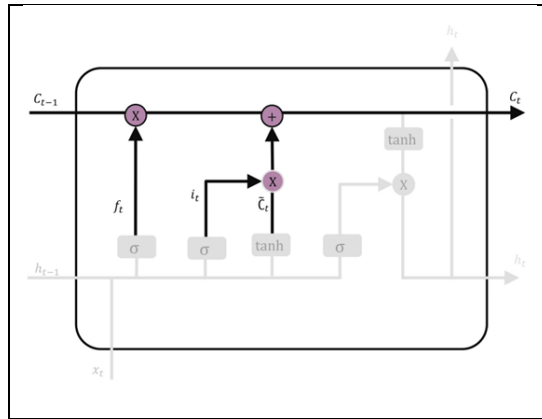


Figure 6: The basic architectures of cell state in LSTM (imtiaz & qusay, may 2022)

To update the cell state, the output of the input gate (i_t) is multiplied with the new memory candidate ($\tilde{C}_t$), and the result is added to the retained information from the previous cell state (C_{t-1}). This process integrates how much old information to forget and how much new information to store (Saurav, Avesh, Sharma, & Hossain, 2023) (imtiaz & qusay, may 2022). The updated cell state (C_t) is computed using Equation (4)

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t \quad (4)$$

2.4.1.4 output gate Layer

The **output gate**, shown in Figure 7, determines the final output of the LSTM cell. This output is derived from the current cell state but in a simplified form. In this layer, the previous hidden state (h_{t-1}) and the current input (x_t) are passed through a sigmoid function (Saurav, Avesh, Sharma, & Hossain, 2023) (imtiaaz & qusay, may 2022), as described in Equation (5).

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (5)$$

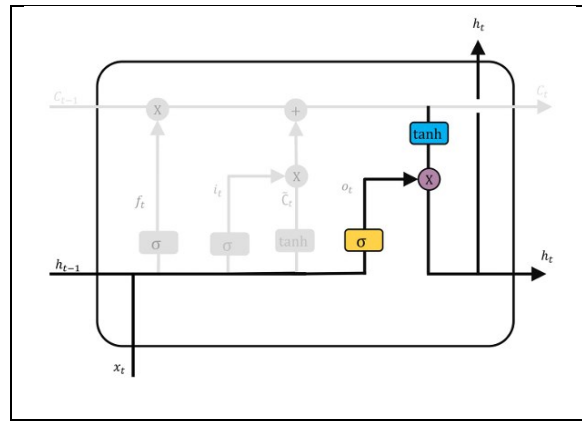


Figure 7: Information processed at Output gate (imtiaaz & qusay, may 2022)

Simultaneously, the updated cell state (C_t) is processed through a tanh function. Next, the result from the tanh function is multiplied by the output of the sigmoid gate, as shown in Equation (6). This multiplication determines what information will be included in the next hidden state (h_t), ensuring that only the selected values are passed forward. The updated cell state (C_t) and hidden state (h_t) are then carried over to the next time step for further processing. This process ensures efficient and selective retention of information across sequences (Saurav, Avesh, Sharma, & Hossain, 2023) (imtiaaz & qusay, may 2022).

$$h_t = o_t \times \tanh(C_t) \quad (6)$$

2.4.2 LSTM Parameters

In training neural networks, selecting and tuning model parameters is essential for achieving optimal performance. Key parameters such as learning rate, batch size, number of layers, drop-out regularization, and loss functions directly impact the model's ability to learn, generalize,

and adapt to various tasks. This project explores these parameters in depth, highlighting their roles, trade-offs, and strategies for effective optimization.

2.4.2.1 Hyperparameters for Scaling: Optimizing Learning Rate and Batch Size

The learning rate, a key hyperparameter in training artificial neural networks, ranges from 0.0 to 1.0 and significantly affects model adaptation. A lower learning rate ensures precise weight updates, improving accuracy but prolonging training and increasing the risk of getting stuck in local minima. Conversely, a higher learning rate accelerates training but risks overshooting optimal solutions (Amrul, Bellina, & Syafaa, 2024). Similarly, larger batch sizes reduce training time but can also cause the model to get stuck in local minima. Striking a balance between learning rate, batch size, **training time**, and accuracy is crucial, and dynamically adjusting these parameters during training can enhance performance but increases model complexity (Yang, et al., 2019).

2.4.2.2 Layers

"Layers" in a neural network consist of computational units or "cells" that process input data and pass information through the network. In LSTM models, layers can be organized in various architectures—such as stacked, bidirectional, or multidimensional—to enhance the network's ability to handle diverse data processing tasks. Stacking multiple LSTM layers allows the model to learn more complex patterns and capture long-term dependencies. However, increasing the number of layers also raises computational complexity and the risk of overfitting, especially if the task is simpler than the model's capacity. While a single layer might lack sufficient complexity, too many layers can lead to performance issues. Achieving optimal performance requires carefully balancing model depth with task complexity and fine-tuning to avoid overfitting while ensuring the model has adequate capacity (Yu, Si, Hu, & Zhang, 2019).

2.4.2.3 Dropout

Dropout is a regularization technique used to prevent overfitting by randomly setting certain nodes in a neural network to zero during training. This forces the model to rely less on specific features and improves generalization by ensuring that training impacts the model as a whole rather than specific nodes. In recurrent neural networks (RNNs), traditional dropout can disrupt learning in long sequences due to accumulated noise. To address this, variational dropout applies a consistent dropout mask across all time steps, including inputs, outputs, and recurrent layers. This Bayesian-inspired approach stabilizes training, effectively regularizes connections,

and is particularly suitable for tasks with large feature sets and limited data, where overfitting is a concern. Dropout layers are added to the model, applying dropout to outputs from the previous layer before passing them to the next, enhancing the model's robustness and reducing the risk of overfitting (Gal & Ghahramani, 2016).

2.4.2.4 Loss Function

The loss function plays a crucial role in determining how the model updates its weights and affects overall performance. For models evaluated using Root Mean Squared Error (RMSE), selecting a loss function that closely aligns with this metric can lead to better optimization and accuracy. RMSE emphasizes larger errors due to its quadratic nature, making it particularly suitable for tasks where significant deviations must be penalized more heavily. By optimizing the model with a loss function compatible with RMSE, training can effectively focus on minimizing these deviations, ensuring higher alignment with the evaluation objective (Rymarczyka, Kuliszc, & Kłosowski, 2023)

2.5 Yahoo Finance as Data Source

Yahoo Finance is a widely used financial platform in the U.S., catering to over 30 million daily retail investors with comprehensive data on publicly traded companies, including stock prices, news, analyst estimates, and financial statements (Bordino, Kourtellis, Laptev, & Billawala., 2014). It plays a pivotal role in market prediction, particularly during earnings announcements, as search data on the platform serves as a proxy for investor attention, correlating with both immediate and long-term stock price movements. Research shows that higher search activity during earnings announcements often results in stronger price reactions and sustained returns, making Yahoo Finance an effective tool for gauging market sentiment. Its predictive power surpasses metrics like Google Trends and SEC EDGAR searches for earnings responses and stock performance. The Yahoo Finance API provides extensive historical datasets of stock prices and financial information for major indices like the S&P 500 and Nasdaq, as well as smaller companies, formatted as CSV files containing columns such as date, open, high, low, close, and trading volume (Lawrence, Ryans, Sun, & Laptev, July 2016)(Bordino, Kourtellis, Laptev, & Billawala., 2014).

2.6 Related Works and Research Gap

In any field of research or innovation, building upon the work of predecessors is not only a time-honored tradition but also a critical step towards progress.

Dey et al. examine the application of recurrent neural networks (RNNs), including advanced variants like Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), for stock price prediction. Highlighting the limitations of traditional machine learning methods in handling complex, non-linear, and time-dependent financial data, the study emphasizes RNNs' ability to model temporal dependencies and automatically extract features. While basic RNNs face challenges like vanishing gradients, LSTM and GRU models address these with mechanisms such as forget and reset gates, improving performance on long-term dependencies. Using stock price data from Honda (low fluctuation), Oracle (moderate fluctuation), and Intuit (high fluctuation) over two decades, the research shows that GRU often outperforms other models due to its simpler architecture and efficient gradient handling, particularly for datasets with high volatility. The study provides a comparative analysis of RNN-based models, offering insights into their strengths and limitations, and sets the stage for further advancements in financial forecasting (Dey, et al., August 2022).

Yenireddy, et al. (Yenireddy, Narayana, Ganesh, Kumar, & Venkateswarlu, September 2024) explore stock market index prediction using Long Short-Term Memory (LSTM) models, addressing the challenge of accurately forecasting stock prices due to their chaotic, nonlinear, and volatile nature. The study aims to predict the closing price of the S&P 500 index by leveraging Yahoo Finance APIs to retrieve historical stock data, which is preprocessed for model training. The methodology incorporates data denoising, feature scaling, and hyperparameter tuning to optimize LSTM performance. Comparative analysis demonstrates that LSTM models outperform traditional methods like Linear Regression and Support Vector Machines, achieving the lowest RMSE (0.42), MAE (0.09), and highest linear correlation (0.99). The results reveal high overlap between actual and predicted prices, highlighting the robustness of LSTM for stock price prediction, even under market anomalies. The study concludes by emphasizing LSTM's superior predictive accuracy and resilience, suggesting its utility for informed investment decisions.

Jarrah et al. (Jarrah & Derbali, 2023). This study focuses on predicting the Saudi Stock Market Index using multivariate time series analysis with a deep learning-based approach, specifically employing Long Short-Term Memory (LSTM) networks. The primary challenge lies in addressing the inherent volatility and unpredictability of stock markets influenced by various economic and financial variables. The researchers propose a solution utilizing historical stock data—closing, opening, highest, and lowest prices—combined with exponential smoothing for

noise reduction and a sliding window technique for data transformation. Two models, univariate and multivariate LSTMs, were implemented, trained on data from the Saudi Stock Exchange (Tadawul), and evaluated. The multivariate model outperformed the univariate model, achieving higher prediction accuracy (97.49% vs. 92.19%), showcasing the efficacy of leveraging diverse data inputs and advanced deep learning methods for precise stock market forecasting.

Charan et al. (Charan, Rasool, & Dubey, January 2022) proposed methods of forecasting of Tata Motors' stock closing prices using four machine learning models: Multivariate Linear Regression, Facebook Prophet, ARIMA, and Long Short-Term Memory (LSTM). The objective is to evaluate the efficiency of these models in predicting stock prices based on historical data obtained from Yahoo Finance, spanning 2004 to 2021. The study addresses the problem of high non-linearity and dynamic nature of stock markets, which make prediction challenging. Each model's performance was measured using Root Mean Square Error (RMSE) as a metric. Data preprocessing involved creating variables from daily stock attributes like open, high, low, and close prices. Results showed that the Prophet model with logistic regression achieved the lowest RMSE (0.1884), followed by LSTM, Linear Regression, and ARIMA, highlighting Prophet's robustness with large datasets and LSTM's advantage in handling short-term patterns. This work underscores the importance of selecting appropriate models for stock market forecasting tasks.

Understanding the existing body of knowledge, methodologies, and breakthroughs is crucial for identifying gaps, refining approaches, and advancing the boundaries of the chosen domain. In the following section, we explore the methodology employed in this study to achieve the previously outlined objectives.

3 Methodology

This study employed a systematic strategy to build and evaluate a Long Short-Term Memory (LSTM) model for predicting stock market changes. The technique comprises five main stages: Data collection, EDA analysis, data pre-processing, LSTM model development and evaluation.

3.1 Data Collection

This project uses a time-series model to predict future stock prices based on historical data, focusing on Germany's automotive sector, one of its fastest-growing industries. It analyzes

stock prices of three major German manufacturers: Mercedes-Benz, BMW, Porsche and Volkswagen.

Rank	Name	Market Cap	Price	Today	Price (30 days)	Country
8	 Porsche P911.DE	\$56.80 B	\$62.35	▼ 2.35%		 Germany
9	 Mercedes-Benz MBG.DE	\$56.43 B	\$58.61	▲ 0.36%		 Germany
10	 BMW BMW.DE	\$50.48 B	\$81.50	▲ 0.49%		 Germany
11	 Volkswagen VOW3.DE	\$49.94 B	\$97.78	▼ 0.45%		 Germany

Figure 8: Global ranking for automotive companies on the Companies Market Cap platform (Companies market cap, visited on 28.12.2024).

According to the German Companies Market Cap platform, as shown in Fig. 6, Porsche ranks 8th with a market capitalization of \$56.56 billion and Mercedes-Benz ranks 9th with a market capitalization of \$56.43 billion and a stock price of \$58.61 (up 0.36%). BMW ranks 10th at \$50.48 billion and \$81.50 (up 0.49%), while Volkswagen (VOW3.DE) ranks 11th at \$49.94 billion (Companies market cap, visited on 28.12.2024). These rankings emphasize Germany's strong position in the global automotive market.

The dataset, spanning from January 1, 2015, to January 30, 2025, was sourced from Yahoo Finance using the yfinance API, a Python library for efficient market data retrieval. The data includes six key columns: Adjusted Close, Close, High, Low, Open, and Volume (table 2). Data storage, loading, and processing were handled using the Pandas library. Newly gathered data was resulting in a total of approximately 15,300 data points collected over the project's timeframe

Date	Adj Close	Close	High	Low	Open	Volume
2024-12-18	78.000000	78.00000	79.19999	78.000000	79.000000	1023527
2024-12-19	77.279999	77.27999	78.37999	76.900002	77.080002	1297262
2024-12-20	77.580002	77.58000	77.86000	76.120003	76.699997	2000177
2024-12-23	76.940002	76.94000	77.68000	76.260002	77.620003	642208

Tabel 2: BMW Dataset from Yahoo Finance.

The data was divided into three sets using a 70/30 split: the first 70% was used for training, while the remaining 30% was reserved for testing and evaluating the LSTM model.

Additionally, a validation set comprising data from the last month was created to mitigate overfitting.

3.2 Exploratory Data Analysis (EDA)

This Exploratory Data Analysis (EDA) (Chatfield, 1986) aims to provide a comprehensive understanding and interpretation of data sets by uncovering inherent patterns and trends through visualization and statistical analysis. EDA is a critical step in data science that facilitates exploration of relationships between features, and lays the groundwork for problem definition and hypothesis formulation. In this analysis, we will explore various aspects of the data, emphasizing risk analysis based on historical performance. The insights gained from this EDA will guide us in selecting the most promising stock among the three for predicting future prices. (Chatfield, 1986)

3.2.1 Closing prices of related stocks over time

The global stock market is heavily influenced by economic events, industry performance, and external shocks such as pandemics. The automotive industry, comprising major players like Volkswagen, Mercedes-Benz, Porsche and BMW, has shown varying levels of resilience and volatility over the years. Stock prices for these companies reflect not only their financial health and market confidence but also broader macroeconomic trends.

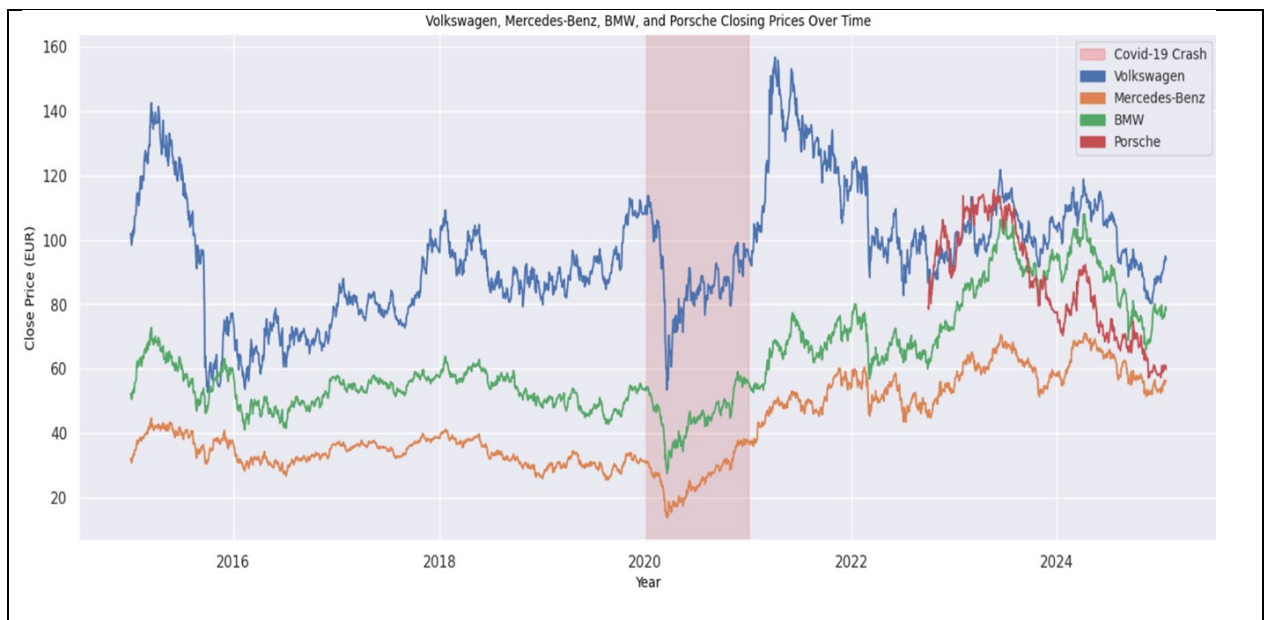


Figure 9: Volkswagen, Mercedes-Benz, BMW, and Porsche Closing Prices Over Time

As illustrated in the Figure 7, the graph tracks the closing prices (in EUR) of these three automotive giants from 2015 to 2025. Volkswagen (blue line) exhibits significant volatility, with notable peaks exceeding 140 EUR, followed by sharp declines, especially during the red-shaded period marking the COVID-19 pandemic crash in 2020. Mercedes-Benz (orange line) and BMW (green line) display relatively lower volatility, maintaining a narrower range of price fluctuations. However, all three companies experienced a downturn during the COVID-19 period, with Volkswagen showing the most pronounced impact. Historical data for Porsche before 2022 is unavailable, therefore, the rest of this study will focus on other automaker brands

3.2.2 Moving average of the various stocks

Moving averages are statistical tools used in financial analysis to smooth out fluctuations in stock prices by calculating the average of an asset's adjusted closing prices over a specified number of days. For example, a 50-day moving average considers the average price over the most recent 50 trading days. Adjusted closing prices account for corporate actions like stock splits and dividends, ensuring accuracy in long-term analysis. Moving averages are integral to stock price prediction because they reveal trends, filter out noise from random price movements, and identify support and resistance levels. They are widely used in technical analysis to signal potential buy or sell opportunities, helping investors make informed decisions (Murphy, 1999).

Figure.10 shows the moving averages of selected German automaker stocks: Volkswagen (VOW3.DE) and Mercedes-Benz (MBG.DE). For Volkswagen, the price fluctuates between €80 and €120, with noticeable peaks around mid-2023 and consistent dips towards late 2024. Mercedes-Benz displays a narrower price range, hovering between €45 and €70. Its trends align closely with the moving averages, showing tighter price convergence over longer periods. The 50-day moving average provides the smoothest representation of overall trends. The price trends for BMW are somewhat similar to Volkswagen, ranging between €60 and €110, with notable peaks around mid-2023. Like the other stocks, BMW's moving averages smooth out short-term fluctuations, with the 50-day moving average indicating broader trends. However, the volatility appears lower than Volkswagen, with more consistent alignment between the adjusted close price and the shorter moving averages.

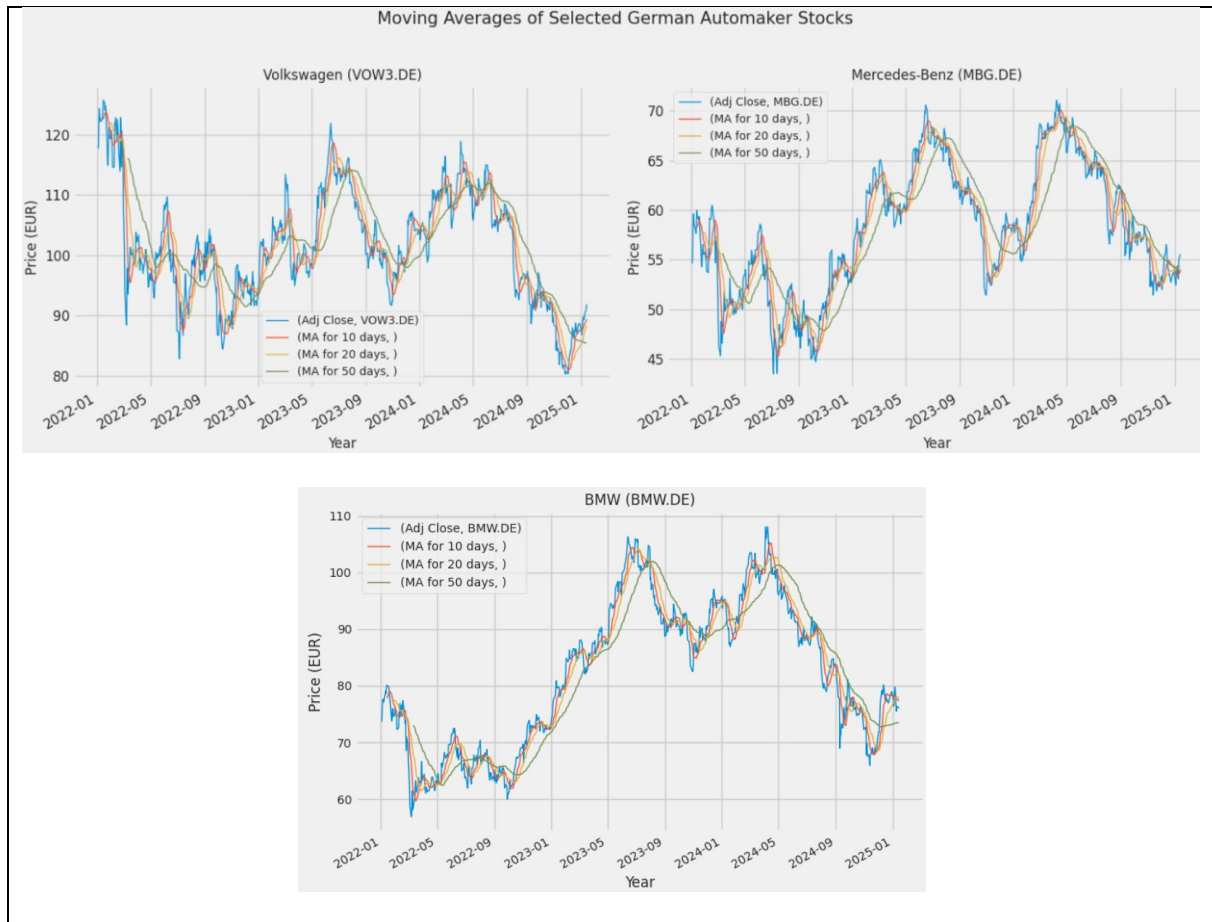


Figure 10: Moving Averages periods of 10, 20 and 50 days of Selected German Automaker Stocks: Volkswagen, Mercedes-Benz and BMW

However, Volkswagen demonstrates the highest price volatility, suggesting higher risk and potentially higher rewards for investors. In contrast, Mercedes-Benz exhibits a more stable price pattern with fewer extreme swings, appealing to conservative investors. BMW lies between the two, balancing moderate volatility with predictable price trends.

3.2.3 daily return of the stock on average

In order to analyze the risk of the stock, we'll need to take a closer look at the daily changes of the stock, and not just its absolute value. We will use pandas to retrieve daily returns for for each stock. As shown in figure.2, The daily returns for Volkswagen appear to fluctuate significantly, with values ranging between approximately -0.075 and 0.1. Spikes in both positive and negative returns suggest periods of high volatility, possibly driven by events such as regulatory developments, or market sentiment. While most of the returns cluster near zero, the notable outliers indicate days with substantial movements in stock price.

for Mercedes-Benz a narrower range of fluctuations compared to Volkswagen, staying within roughly -0.06 and 0.08. This indicates relatively lower volatility. The clustering of data near zero highlights a pattern of stable daily movements, with fewer extreme outliers. The steadiness may reflect consistent performance or investor confidence in the company.

BMW (BMW.DE): The daily returns for BMW reveal an intermediate volatility level, with values ranging approximately between -0.1 and 0.075. While the returns are mostly concentrated near zero, there are notable negative outliers in late 2024, suggesting significant downward movement, potentially due to macroeconomic or company-specific factors.

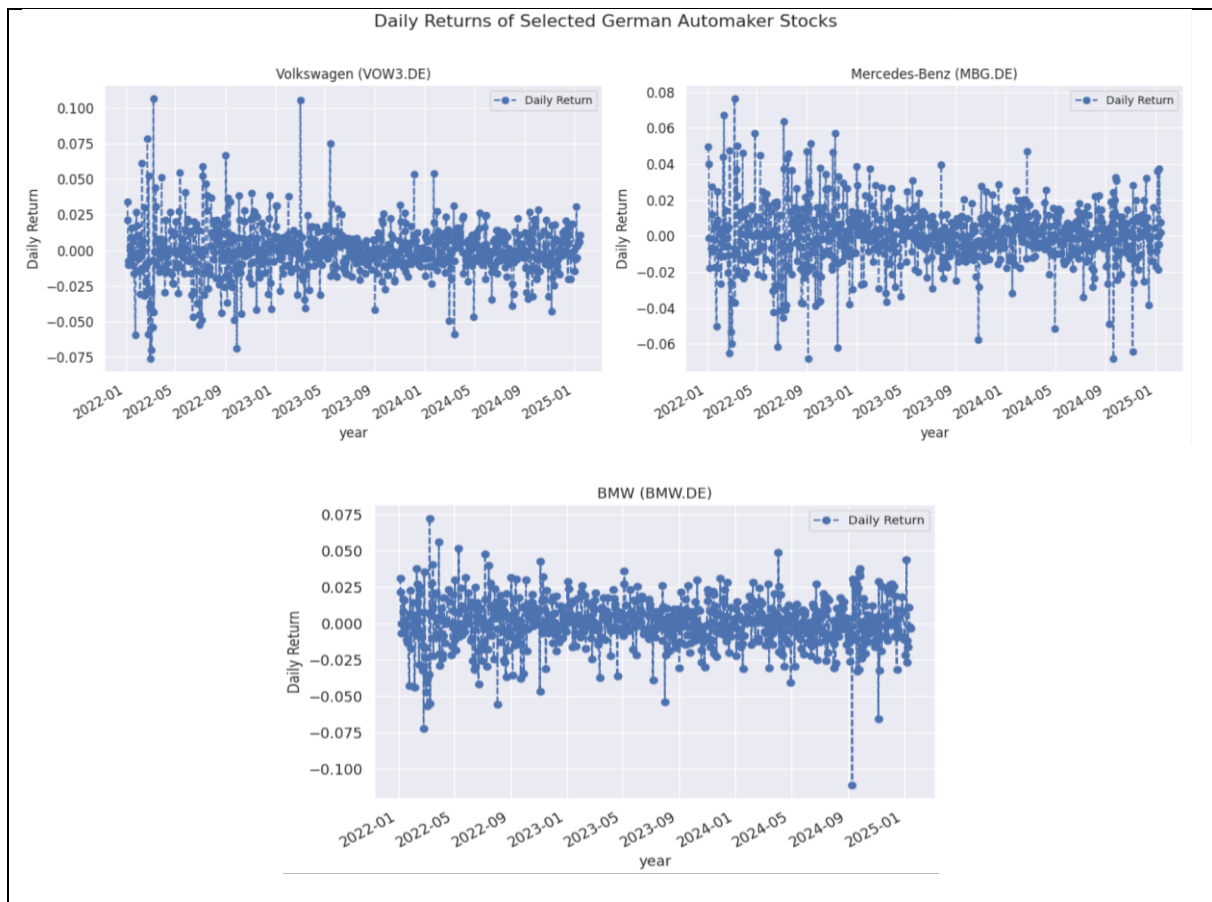


Figure 11: Daily Returns of Selected German Automaker Stocks

These graphs underscore different risk-return profiles for the stocks. Volkswagen's significant volatility implies it is influenced more by external shocks or internal news. Mercedes-Benz's steadier performance may highlight its robust operations and investor confidence. BMW balances between volatility and stability, making it a moderate-risk option compared to the other two.

3.2.4 risk of investing in a particular stock

In stock prediction, understanding the relationship between risk and return is essential for selecting stocks that are more predictable and align with forecasting objectives. A risk-return plot provides a visual representation of this relationship, helping to identify stocks with patterns that are better suited for predictive models. Figure 12 is a risk-return plot showing the relationship between the expected return and risk for selected stocks. Among these, BMW.DE stands out as the most attractive, offering the highest expected return (approximately 0.00015) with the lowest risk (around 0.01675). MBG.DE has slightly higher risk (about 0.0175) but delivers a lower expected return (~ 0.0001), making it less efficient in terms of risk-adjusted returns. In contrast, VOW3.DE exhibits the highest risk (0.0185) and a slightly negative expected return (-0.00005), suggesting it is the least favorable option for investors focused on optimizing risk and return.

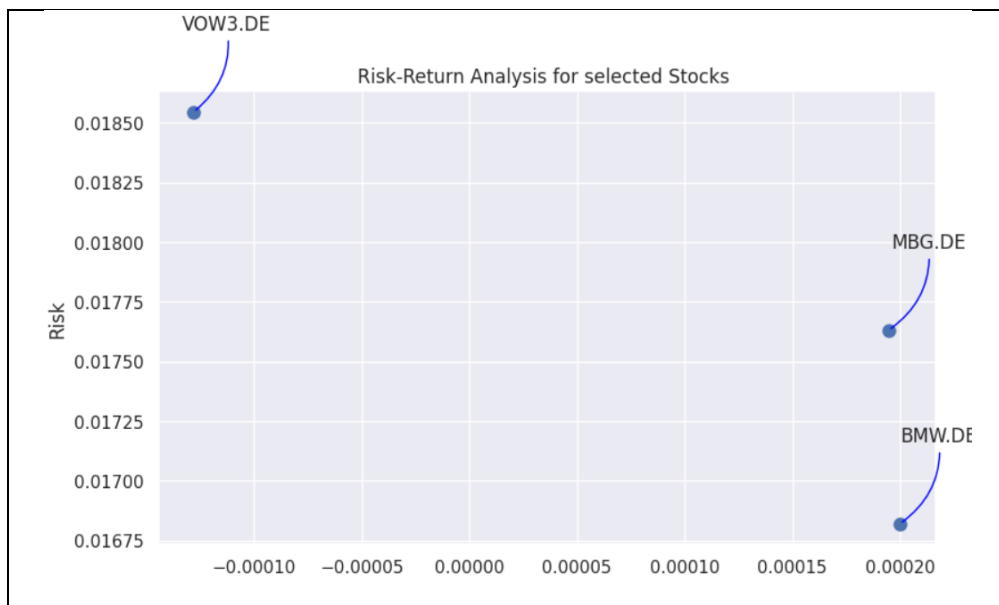


Figure 12: Risk-Return Analysis for selected automaker Stocks

BMW.DE is the most suitable candidate for an LSTM-based stock prediction model due to its lower volatility and positive expected return. LSTM models excel at capturing time-series patterns, and BMW.DE's smoother and more predictable behavior makes it easier to model compared to the higher volatility and negative return patterns of VOW3.DE.

3.3 Data Preprocessing

3.3.1 Data Cleaning

Before proceeding with the analysis, we perform data cleansing and filtering to ensure the dataset's quality and integrity. This step involves removing any null values, correcting formatting issues, and replacing zero values in a specified DataFrame column with the mean of the adjacent

non-zero values using Pandas' rolling mean and boolean indexing for efficiency. The process handles edge cases effectively and creates a copy of the original DataFrame to preserve data integrity. After cleansing, the data is organized into Pandas DataFrames for each stock, facilitating easier manipulation and analysis.

3.3.2 Normalization

Normalizing data enhances model training by accelerating convergence, reducing training time, and preventing issues such as excessively large gradients (Hochreiter & Schmidhuber, 1997), (Kumar, 2021). To achieve this, data was scaled to a range of 0 to 1 using the MinMaxScaler from the sklearn library. Without normalization, features with larger numerical ranges dominate the training process, which is undesirable.

Normalization ensures that features with vastly different ranges contribute equally during training. For example, the "Volume" feature for BMW.DE stock can exceed 10,900,000, while other features, rarely exceed 120. If these features were normalized together, smaller features would compress near 0, and "Volume" would dominate near 1, rendering the smaller features insignificant. To address this imbalance, a logarithmic transformation was applied to the "Volume" feature, reducing its scale and aligning it more closely with the ranges of other features.

Conversely, features OHLC prices share an inherent relative relationship that must be preserved. Normalizing these features independently could distort their relationships, potentially resulting in unrealistic scenarios, such as "Low" values appearing larger than "High" or "Open" values. To avoid this, these features were normalized collectively, ensuring their relative differences remained intact and meaningful. This approach preserves the integrity of the price data (Atsalakisa & Valavanis, 2009).

3.4 Model Parameters

This section outlines the selection process for specific hyperparameters and describes the approach used for their tuning. A comprehensive analysis of the parameters and features will be provided in the 'Results and Discussion' section.

3.4.1 Learning Rate and Batch Size

In this project, the learning rate and batch size were chosen to optimize training efficiency and model performance. The model uses the Adam optimizer, which has a default learning rate of

0.001. While this value generally provides a good balance between convergence speed and stability, it can be adjusted if needed to refine training dynamics.

A batch size of 3 was selected for training. Although smaller than typical values, this decision was made to improve accuracy and adapt to the dataset's characteristics. Smaller batch sizes provide noisier gradient updates, which can help escape local minima and improve generalization. The use of early stopping ensured that training stopped once the validation loss stopped improving, preventing overfitting while maintaining efficient training.

3.4.2 Layers

The architecture used in this project consists of three LSTM layers, each with 50 units. This configuration was chosen based on experiments optimizing the number of layers for this application. The three-layer structure was found to provide the best balance between capturing the temporal dependencies in the data and maintaining computational efficiency. Each LSTM layer uses the `return_sequences` parameter to enable stacking, with the final LSTM layer omitting this parameter to produce the required output shape. The inclusion of multiple LSTM layers allows the model to learn hierarchical representations of the data, where the lower layers capture short-term dependencies and the higher layers capture more abstract patterns. This architecture is particularly suited to the time-series nature of stock price prediction.

3.4.3 Dropout

A dropout rate of 30% was applied to each LSTM layer to mitigate overfitting by randomly deactivating neurons during training. This prevents the model from becoming overly reliant on specific features and ensures better generalization to unseen data. Dropout rates of up to 50% and configurations without dropout were tested, but 30% was found to strike an optimal balance between reducing overfitting and preserving model capacity.

By incorporating these design choices, the model is robust to overfitting, effectively captures the temporal structure of the data, and maintains computational efficiency suitable for the project's scope.

3.4.4 Window Size

In this project, a window size of 60 was used to create sequences of historical stock data as input for the model. The window size represents the length of the input sequence provided to

the LSTM model during training and evaluation. This choice allows the algorithm to capture patterns in the data over a meaningful time span while maintaining a balance between model complexity and computational efficiency. A shorter window size might miss longer-term trends, whereas a larger window size could increase the risk of overfitting and result in longer training times. For a stock prediction task using daily data, this window size of 60 corresponds to approximately two months of historical data, which was deemed sufficient to recognize relevant trends and patterns for future forecasting.

3.4.5 Number of Cells

The LSTM model utilized 50 cells per layer, with three stacked LSTM layers in total. The number of cells in each layer determines the capacity of the model to learn complex patterns in the data. During model design, 50 cells were chosen as a compromise between performance and computational efficiency. Larger numbers of cells might improve accuracy but could lead to overfitting or significantly longer training times. The stacked architecture, with dropout layers between each LSTM layer, ensures that the model learns hierarchical temporal features while reducing overfitting risks. This architecture was also optimized to handle multivariate input data effectively.

3.4.6 Model Callbacks

To optimize the training process, the EarlyStopping callback from Keras was implemented. The early stopping mechanism monitored the validation loss during training and stopped the training process once the validation loss failed to improve for 10 consecutive epochs (patience=10). This approach prevented overfitting and ensured that the model parameters corresponding to the lowest validation loss were retained. The use of early stopping reduced unnecessary training time and ensured that the model converged to an optimal solution. Additionally, the best-performing model weights were restored after training, providing a robust baseline for making predictions on unseen data.

4 Results and Discussion

This section presents the results of the stock price forecasting model and discusses its performance. Tables and figures are included to provide a detailed analysis of the findings. The model used is a three-layered LSTM network optimized for the given dataset of BMW stock prices. Several hyperparameters, such as the number of LSTM units and dropout rate, were tested to improve model performance. Early stopping was implemented to prevent overfitting.

4.1 Model Evaluation

The LSTM model was trained and tested using a normalized dataset of historical BMW stock prices. The model was evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) metrics. The results are summarized in Table 3.

Metric	Training	Testing	Validation
RMSE	1.37	1.38	1.08
MAE	0.95	0.95	0.86

Tabel 3: evaluation metrics for LSTM Model

Table 3 shows that the RMSE and MAE values for the training, testing, and validation sets are within acceptable ranges, indicating that the model is learning the underlying patterns effectively. However, the slightly higher values in the testing set suggest minor overfitting.

4.2 Predictions vs. Actual Values Across Training, Testing, and Validation Sets

Figure 13 illustrates the comparison between the predicted and actual stock prices for the validation set. The predicted values closely follow the actual prices, demonstrating the model's ability to capture trends.

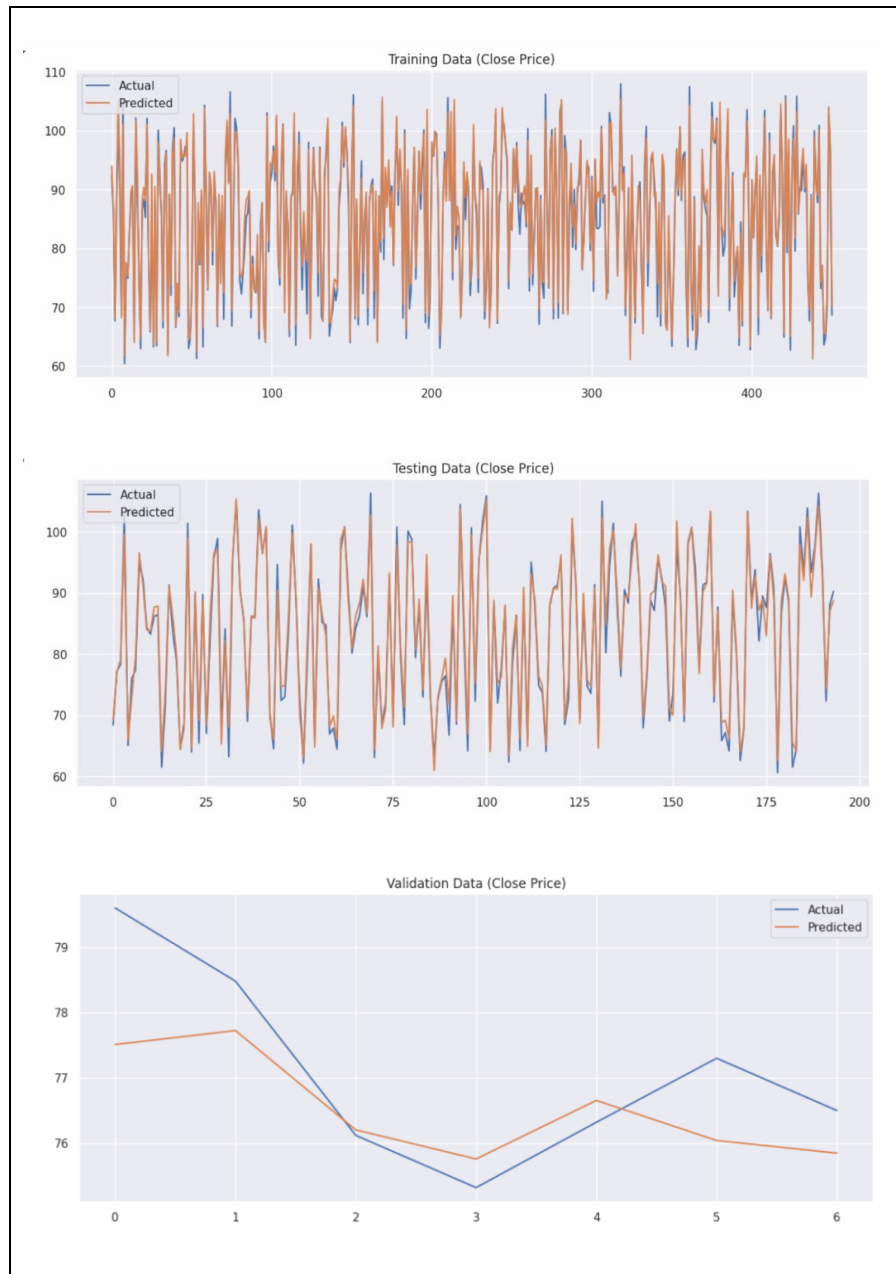


Figure 13: Close-Up View of Predicted vs. Actual BMW Stock Prices for Training, Testing, and Validation Sets.

4.3 Impact of Hyperparameters

The model's performance was analyzed by varying key hyperparameters, such as the number of LSTM units and window size. Table 2 provides a summary of the results for different configurations.

LSTM Units	Window Size	Validation RMSE	Validation MAE
50	60	2.46	1.30
64	60	2.55	1.45
128	60	2.65	1.44

Tabel 4: Performance Comparison of LSTM-Models with Different Units and Fixed Window Size

The best performance was achieved with 64 LSTM units and a window size of 60. Increasing the number of units beyond this point resulted in higher RMSE, possibly due to overfitting.

4.4 Future Price Forecasting

The model was also used to predict stock prices for the next month. The predicted values are as follows:

Day	Predicted Price
03.02.25	77.30
04.02.25	77.03
05.02.25	76.76
06.02.25	76.49
07.02.25	75.98

Tabel 5: Predicted Prices Over Time

While these predictions provide a general idea of the stock's short-term trajectory, the directional accuracy is limited, as observed in similar studies.

4.5 Discussion

The results highlight the effectiveness of using LSTM models for time series forecasting. The model successfully captures short-term trends but shows sensitivity to hyperparameters and data preprocessing. Normalizing the data and incorporating technical indicators, such as SMA and RSI, improved the model's learning capacity. However, challenges remain in achieving higher directional accuracy, especially during volatile market conditions.

A notable observation was the effect of data trends on predictions. Periods of uptrends or downtrends influenced the model's ability to generalize, leading to occasional over- or underestimations. Future work could explore dynamic learning rates and additional feature engineering to address these issues.

In conclusion, the LSTM model demonstrated satisfactory performance in forecasting BMW stock prices. Further optimization and the inclusion of additional data, such as macroeconomic indicators, could enhance its predictive accuracy.

Conclusion

This study has demonstrated the effectiveness of Long Short-Term Memory (LSTM) networks in predicting stock market trends, specifically focusing on the German automotive sector. The model successfully captured short-term fluctuations and provided valuable insights into stock price movements. Our evaluation using RMSE and MAE metrics indicated that the LSTM model was able to learn underlying patterns from historical data, though some sensitivity to hyperparameters and market volatility was observed.

Despite the promising results, challenges remain. The model's performance is influenced by external economic conditions, corporate events, and unexpected market shifts, which are difficult to capture using purely historical data. Future research could explore integrating additional macroeconomic indicators, such as inflation rates, interest rates, and global economic trends, to enhance predictive accuracy. Furthermore, experimenting with hybrid models, combining LSTM with other deep learning architectures like Transformers or attention-based mechanisms, may improve long-term forecasting capabilities. While LSTM models provide a strong foundation for stock market prediction, further refinements and hybrid approaches could significantly enhance their robustness and applicability in dynamic financial environments.

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